UNIVERSITY OF TORONTO
Faculty of Arts & Science

April 2024 Examinations

CSC 384 H1S

Duration: 3 hours

Aids Allowed: Two *double*-sided (printed or handwritten) *8.5"×11"* aid sheets.

*Do **not** turn this page until you have received the signal to start.*
In the meantime, write your name, student number, and UTORid below (please do this now!) and *carefully* read *all* the information on the rest of this page.

Given Name(s):

Family Name(s):

Student Number (9 or 10 digits):

UTORid (e.g., `pitfra12`):

- Turn off and place all cell phones, smart watches, electronic devices, and unauthorized study materials in your bag under your desk. If it is left in your pocket, it may be an academic offence.
- When you are done your exam, raise your hand for someone to come and collect your exam. *Do not collect your bag and jacket before your exam is handed in.*
- If you are feeling ill and unable to finish your exam, please bring it to the attention of an Exam Facilitator so it can be recorded before leaving the exam hall.
- In the event of a fire alarm, do **not** check your cell phone when escorted outside.
- This final examination consists of 8 question on 18 page (including this one), printed on both sides of the paper. *When you receive the signal to start, please make sure that your copy of the examination is complete.*
- Answer each question directly on the examination paper, in the space provided, and use a "blank" page for rough work. If you need more space for one of your solutions, use one of the "blank" pages and *indicate clearly the part of your work that should be marked.*
- *Remember that, in order to pass the course, you must achieve a grade of at least **40%** on this final examination.*
- As a student, you help create a fair and inclusive writing environment. If you possess an unauthorized aid during an exam, you may be charged with an academic offence.

Marking Guide

Nº 1:	_____/10
Nº 2:	_____/ 8
Nº 3:	_____/15
Nº 4:	_____/14
Nº 5:	_____/11
Nº 6:	_____/13
Nº 7:	_____/10
Nº 8:	_____/ 9
TOTAL:	_____/90

STUDENTS MUST HAND IN ALL EXAMINATION MATERIALS AT THE END

Use the space on this "blank" page for scratch work, or for any solution that did not fit elsewhere.
Clearly label each such solution with the appropriate question and part number.

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Question 1. [Short Answer Questions] [10 marks]

Part (a) [2 marks]

In Uniform Cost Search (UCS), as soon as a node satisfying the goal condition is added to the Frontier (or OPEN list) UCS has found an optimal solution. **(True/False)? Explain why.**

Part (b) [3 marks]

In first-order logic name three things that are "terms"?

Part (c) [5 marks]

Let A, B, and C be three discrete random variables. Determine if each of the following equations are true or false. Do not make any assumptions unless specified. You do not need to show any work.

1. $P(A, B, C) = P(A)P(B|A)P(C|A)$.
2. $P(A, B) = \sum_C P(C, B, A)$.
3. if A and B are independent, then $P(A) = \sum_B P(B)P(A)$.
4. if A and B are independent, then A and B are conditionally independent given C.
5. if A and B are conditionally independent given C, then A and B are independent.

Question 2. [Search] [8 marks]

Suppose you want to find a simple path with maximum cost from source to goal. (Recall, a simple path is a path without cycles, i.e., no vertex appears more than once.) To do so, let us assume you have modified the UCS algorithm as described in Algorithm 1, essentially, there is one change introduced in the standard UCS algorithm:

1. At line 15, $\hat{g}[v] = max(\hat{g}[v], \hat{g}[u] + c(u, v))$ instead of $\hat{g}[v] = min(\hat{g}[v], \hat{g}[u] + c(u, v))$.

(Do not worry about the algorithm description, you can simply assume that in UCS, we are using max instead of min for the update of $\hat{g}[v]$). Lets call the modified algorithm as UCS^{max}.

Algorithm 1 UCS^{max} Algorithm

```
1: function FindPathToGoal(s)
2:     F(Frontier)← PriorityQueue()
3:     E(Explored)← ∅
4:     F.push(s)
5:     ĝ[s] ← 0
6:     while F is not empty do
7:         u ← F.pop()
8:         if GoalTest(u) then
9:             return path(u)
10:        end if
11:        E.add(u)
12:        for all children v of u do
13:            if v not in E then
14:                if v in F then
15:                    ĝ[v] = max(ĝ[v], ĝ[u] + c(u, v))
16:                else
17:                    F.push(v)
18:                    ĝ[v] = ĝ[u] + c(u, v)
19:                end if
20:            end if
21:        end for
22:    end while
23:    return Failure
24: end function
```

Prove the following statement:

> The algorithm UCS^{max} will always return the simple path with maximum cost from source to goal if such a path exists.

You can assume all the conditions under which we proved UCS to be optimal, i.e., the graph is finite and for some $\varepsilon > 0$, every edge has a cost $\geq \varepsilon$.

Question 2. (continued)

Question 3. [CSP] [15 marks]

Let us consider the following Constraint Satisfaction Problem (CSP) formulation.

- Variables: $\{A, O, P, R, T\}$.
- Domain of each variable: $\{0, 1, 2, 3, 4, 5, 6, 7, 8\}$.
- Constraints:
  1. No two variables have the same value.
  2. R + P = O
  3. T + T = T
  4. O + O = A

Assume that the variables are assigned alphabetically, that is, in the order $\langle A, O, P, R, T\rangle$ — variable A will be assigned first, O afterwards and so on.

Part (a) [10 marks]

Assume that the domain values of variables are tried in the increasing order, i.e., $\langle 0, 1, 2, 3, 4, 5, 6, 7, 8\rangle$ — value 0 is tried before 1, and so on. Find the values assigned to the variables if Backtracking search with full inference is performed. Please state the values in the table below. You are NOT required to show your work; but if you do, you may get partial marks if your final answer is incorrect.

Variable	Value
A	
O	
P	
R	
T	

Question 3. (continued)

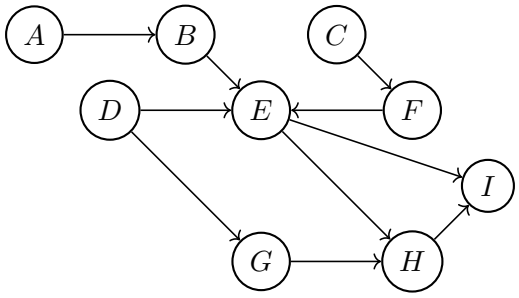
Part (b) [5 marks]

Will the values assigned to variables change if the domain values are tried in the decreasing order, that is, $\langle 8, 7, 6, 5, 4, 3, 2, 1, 0 \rangle$ — value 8 is tried before 7, and so on ? If yes, please state the values in the table below. If no, please explain why.

Variable	Value
A	
O	
P	
R	
T	

Question 4. [Uncertainty Representation] [14 marks]

Consider the Bayesian network shown below.



Part (a) [8 marks]

Let $X \perp Y|S$ denote that X is independent of Y given the evidence set S (note that S can be empty, and in that case we use the notation $X \perp Y$). Circle either **True** or **False** in each of the statements below:

(i) **True** or **False**: $A \perp B$

(ii) **True** or **False**: $A \perp C$

(iii) **True** or **False**: $A \perp D|E$

(iv) **True** or **False**: $A \perp I|E$

(v) **True** or **False**: $B \perp C|I$

(vi) **True** or **False**: $F \perp A|H$

(vii) **True** or **False**: $D \perp I|\{E, G\}$

(viii) **True** or **False**: $C \perp H|G$

Part (b) [3 marks]

Provide a product decomposition of the joint distribution over $A, B, C, D, E, F, G, H, I$, based on the network in Part (a).

$$P(A, B, C, D, E, F, G, H, I) =$$

Part (c) [3 marks]

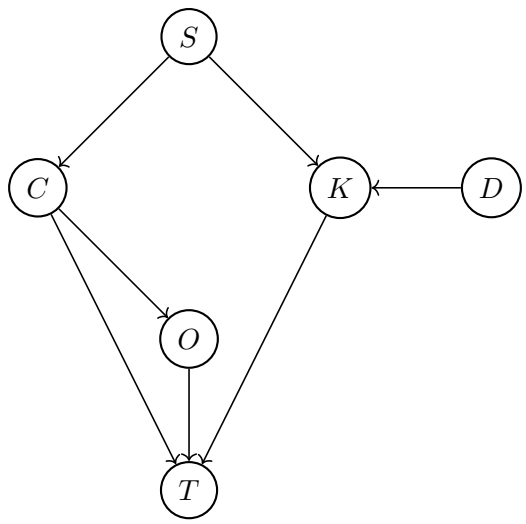
The number of elements in the domain of each variable is shown below. What is the minimum number of values (i.e., the sum of the domain sizes of the factors) that must be stored to recover the joint distribution between $A, B, C, D, E, F, G, H, I$? Show your work.

$A:2$	$B:4$	$C:2$	$D:3$	$E:5$	$F:5$	$G:3$	$H:2$	$I:2$

Question 5. [Uncertainty Inference] [11 marks]

It seems that this test is harder than expected and time is almost up. You decide that rather than working on the questions, your time is best spent computing the probability that you pass the test instead.

You model the probability that you pass the test T via the following Bayesian network.



Here, S denotes whether you studied; C denotes whether you cheat (don't do it); O denotes whether you got caught; D denotes whether the test is difficult; and K denotes whether you know the answers to the questions.

The conditional probability tables for the network are:

$P(s)$
0.8

$P(d)$
0.4

	$P(o\|C)$
c	0.7
$\neg c$	0

	$P(c\|S)$
s	0.2
$\neg s$	0.5

		$P(k\|S,D)$
s	d	0.7
s	$\neg d$	0.9
$\neg s$	d	0.1
$\neg s$	$\neg d$	0.4

			$P(t\|O,C,K)$
o	c	k	0
o	c	$\neg k$	0
o	$\neg c$	k	0
o	$\neg c$	$\neg k$	0
$\neg o$	c	k	1
$\neg o$	c	$\neg k$	0.7
$\neg o$	$\neg c$	k	0.9
$\neg o$	$\neg c$	$\neg k$	0.1

The day before the test, your "friend" found an advanced copy in the professor's office. They offer to give you the answers if you get them bubble-tea. You decided against it because of your morals. Now you want to know the probability that you will pass the test. More specifically, you want to determine $P(t|c, \neg k)$. **Show** your work.

Question 5. (continued)

Question 6. [Knowledge Representation] [13 marks]

Part (a) [7 marks]

Translate the following English sentences into first-order logic:

1. Every mother has a child

2. Every mother is an adult human

3. Some adults are not mothers.

Part (b) [6 marks]

Consider a Knowledge Base Φ that consists of the following first-order sentences:

- $\forall x(P(x) \vee Q(x))$
- $\forall x(P(x) \rightarrow \neg\exists y\, R(y,x))$
- $\forall x(Q(x) \rightarrow \exists y\, R(y,x))$
- $\exists x, y((P(x) \wedge Q(y)) \rightarrow R(y,x))$

Provide **one** model of Φ.

Question 7. [Reasoning] [10 marks]

Part (a) [4 marks]

Convert the following logical formulas to <u>*clausal form*</u>. **Clearly indicate any Skolem functions or constants used in the conversion.**

1. $\forall x(roman(x) \rightarrow loyal_to(x, \textbf{caesar}) \vee hates(x, \textbf{caesar}))$
   Note: caesar is a constant symbol

2. $\forall x(\exists y(person(y) \wedge loyal_to(x, y)))$

Part (b) [6 marks]

Use resolution to prove that $P(x) \wedge Q(y) \wedge R(z)$ is a logical consequence of the following knowledge base:

$$KB = \{R(z) \vee S(t),\ \neg S(t) \wedge P(x),\ Q(y) \vee S(t)\}$$

Present your proof by a **resolution tree**.

Use the space on this "blank" page for scratch work, or for any solution that did not fit elsewhere.
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Question 8. [Game Tree Search] [9 marks]

Consider a game between two players, MIN and MAX. In the game, there are 16 coins and one of them is selected. The utility of a coin is the **expected** number of tosses until **two consecutive heads** occur.

The game tree shown is in Figure 1. The leaf nodes s_{16} to s_{31} in the game tree correspond to unique coins, each of which have their probability of heads shown below the respective leaf nodes. The MAX player will try to maximize the utility and MIN player will try to minimize the utility. In the selection process, MAX player goes first, the square nodes corresponds to the MAX player's turn.

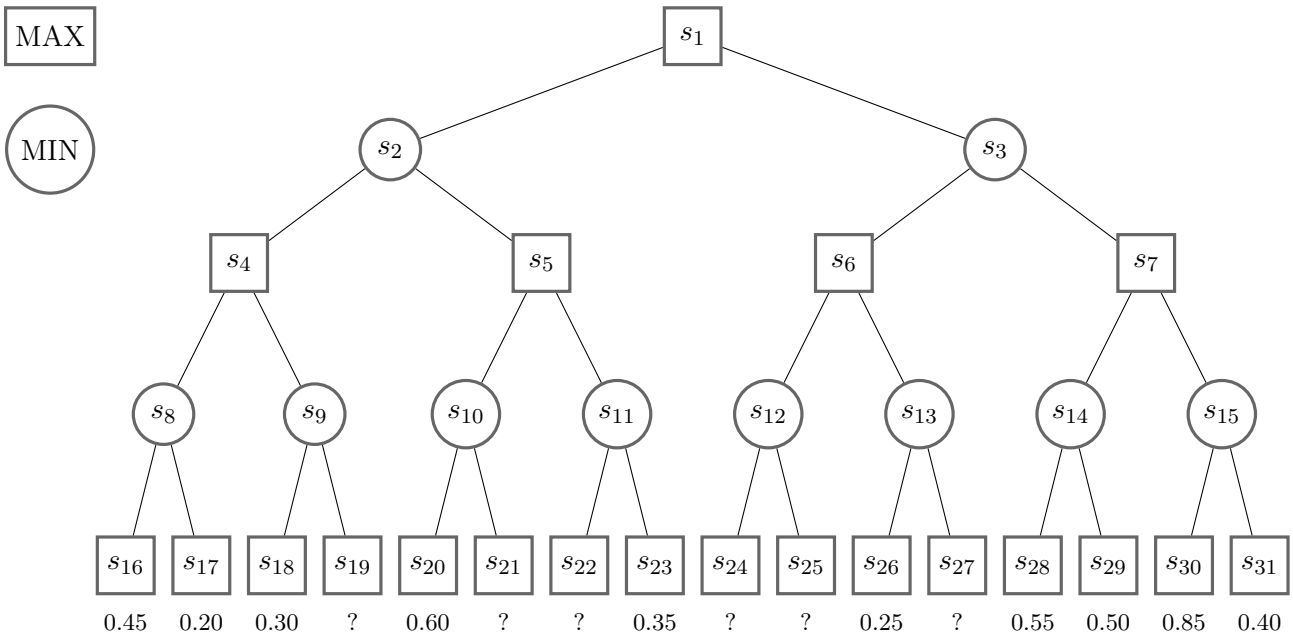

Figure 1: Each coin's probability of getting heads is shown below the corresponding leaf node. The squares correspond to the MAX player's turn and the circles correspond to that of MIN player.

Assume that we run the α-β-algorithm on the game tree in Figure 1 from **left to right**. There are 6 leaf nodes that have yet to be assigned with unique coins, indicated with '?' in Figure 1. The remaining coins are shown in Table 1.

Coin number	$P(heads)$
c_1	0.90
c_2	0.80
c_3	0.75
c_4	0.70
c_5	0.65
c_6	0.15

Table 1: Table of unassigned coins and their probability of heads

Part (a) [6 marks]

Fill in the missing coins in the table below for leaf nodes such that maximum number of leaf nodes are unevaluated (i.e., their evaluation is skipped) by the α-β-pruning algorithm.
Hint: If the probability of getting heads is p, then the expected number of tosses until two consecutive heads occur is equal to $\frac{1+p}{p^2}$. However, you do not really need to calculate the exact values of the expected number of tosses to answer this question.

Leaf node	Coin number
s_{19}	
s_{21}	
s_{22}	
s_{24}	
s_{25}	
s_{27}	

Part (b) [3 marks]

Write down the leaf nodes that are unevaluated (i.e., their evaluation is skipped because the branches containing them are pruned) by the α-β-algorithm. If no leaf nodes are pruned write **None**.

Pruned leaf nodes: _____

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